9 Principles of Tooth Preparations

The design of a preparation for a cast restoration and the execution of that design are governed by five principles:

1. Preservation of tooth structure
2. Retention and resistance
3. Structural durability
4. Marginal integrity
5. Preservation of the periodontium

Preservation of Tooth Structure

In addition to replacing lost tooth structure, a restoration must preserve remaining tooth structure. Intact surfaces of tooth structure that can be maintained while producing a strong, retentive restoration should be saved if patient acceptance and retention requirements will permit it. Whole surfaces of tooth structure should not be needlessly sacrificed to the bur in the name of convenience or speed.

Preservation of tooth structure in some cases may require that limited amounts of sound tooth structure be removed to prevent subsequent uncontrolled loss of larger quantities of tooth structure. This is the rationale for the removal of 1.5 mm of occlusal tooth structure when preparing a tooth for a mesio-occlusodistal (MOD) onlay. The metal on the occlusal surface can protect against dramatic failures, such as fracture of tooth structure, as well as the less obvious failures that may be caused by the flexure of tooth structure.

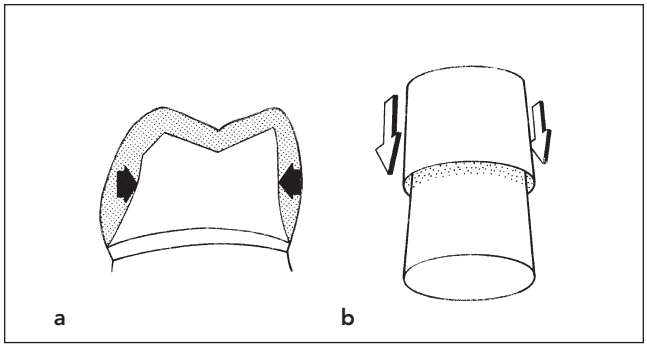


Fig 9-1 An extracoronal restoration *(a)* uses opposing external surfaces for retention *(b)*.

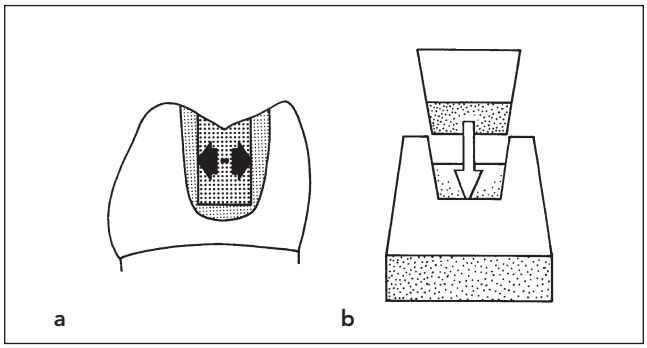


Fig 9-2 An intracoronal restoration *(a)* uses opposing internal surfaces for retention *(b)*.

Retention and Resistance

For a restoration to accomplish its purpose, it must stay in place on the tooth. No cements that are compatible with living tooth structure and the biologic environment of the oral cavity possess adequate adhesive properties to hold a restoration in place solely through adhesion. The geometric configuration of the tooth preparation must place the cement in compression to provide the necessary retention and resistance.

Retention prevents removal of the restoration along the path of insertion or long axis of the tooth preparation. *Resistance* prevents dislodgment of the restoration by forces directed in an apical or oblique direction and prevents any movement of the restoration under occlusal forces. Retention and resistance are interrelated and often inseparable qualities.

The essential element of retention is two opposing vertical surfaces in the same preparation. These may be external surfaces, such as the facial and lingual walls of a full coverage crown (Fig 9-1a). An extracoronal restoration is an example of veneer, or sleeve, retention (Fig 9-1b). The opposing surfaces can also be internal, such as the facial and lingual walls of the proximal box of a proximo-occlusal inlay (Fig 9-2a). An intracoronal restoration resists displacement by wedge retention (Fig 9-2b). Many restorations are a combination of the two types.

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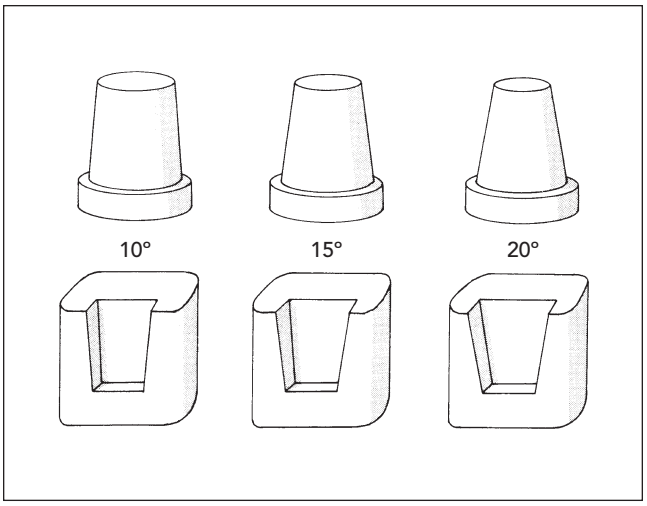


Fig 9-3 External *(top row)* and internal *(bottom row)* opposing surfaces demonstrate tapers of 10, 15, and 20 degrees.

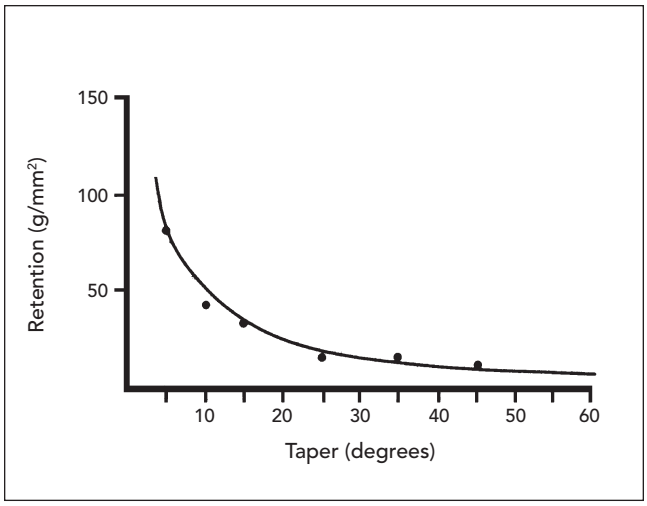


Fig 9-4 As taper increases, retention decreases. (Modified from Jorgensen[4] with permission.)

Taper

Because a cast metal or ceramic restoration is placed on or in the preparation after the restoration has been fabricated in its final form, the axial walls of the preparation must taper slightly to permit the restoration to seat; ie, in the occlusal direction, two opposing external walls must gradually converge or two opposing internal surfaces of tooth structure must diverge (Fig 9-3). The terms *angle of convergence* and *angle of divergence* can be used to describe the respective relationships between the two opposing walls of a preparation.

The relationship of one wall of a preparation to the long axis of that preparation is the *inclination* of that wall. A tapered diamond or bur will impart an inclination of 2 to 3 degrees to any surface it cuts if the shank of the instrument is held parallel to the intended path of insertion of the preparation. Two opposing surfaces, each with a 3-degree inclination, would give the preparation a 6-degree taper.

Theoretically, the more nearly parallel the opposing walls of a preparation, the greater should be the retention. The most retentive preparation should be one with parallel walls. Indeed, parallel walls were advocated by some early authors.[1,2] However, parallel walls are impossible to create in the mouth without producing preparation undercuts. Preparation walls are tapered to allow their visualization, prevent undercuts, compensate for inaccuracies in the fabrication process, and permit more nearly complete seating of restorations during cementation.

Ward[3] was one of the first to recommend tapering as such, prescribing 5% to 20% per inch (3 to 12 degrees, respectively). Jorgensen[4] and Kaufman et al[5] have demonstrated experimentally that retention decreases as taper is increased (Fig 9-4). Recommendations for optimum axial wall taper of tooth preparations for cast restorations have ranged from 3 to 5 degrees[6] to 6 degrees[7] to 10 to 14 degrees.[8] To minimize stress in the cement interface between the preparation and restoration, a taper of 2.5 to 6.5 degrees has been suggested as optimum. There is only a slight increase in stress as taper is increased from 0 to 15 degrees[9]; however, at 20 degrees, stress concentration was found to increase sharply.

Studies of actual crown preparations have shown average tapers that have been much greater than the values recommended. Ohm and Silness[10] reported mean tapers of 19.2 degrees mesiodistally and 23.0 degrees faciolingually on vital teeth and 12.8 degrees mesiodistally and 22.5 degrees faciolingually on nonvital teeth. Mack[11] found an average clinical taper of 16.5 degrees. Weed et al[12] found that dental students could produce full veneer crown preparations with a taper of 12.7 degrees on typodonts, but their clinical preparations had a mean taper of 22.8 degrees. Noonan and Goldfogel,[13] surveying 909 student-prepared full gold crown preparations, reported an overall mean taper of 19.2 degrees. On proficiency examinations, preparation tapers were decreased by 20%. Dies taken at random from commercial laboratories by Eames et al[14] were found to have an average overall taper of 20 degrees.

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Table 9-1 Optimum degree of tooth preparation taper

Tooth	M/D	F/L	Overall
Maxillary			
Anterior tooth*	10	10	10
Premolar*	14	14	14
Molar*	17	21	19
Isthmus†	NA	NA	7
Box†	NA	NA	7
Mandibular			
Anterior tooth*	10	10	10
Premolar*	16	12	14
Molar*	24	20	22
Isthmus†	NA	NA	12
Box†	NA	NA	12

M/D, mesiodistal; F/L, faciolingual; NA, not applicable.
*Convergence angle.
†Divergence angle.

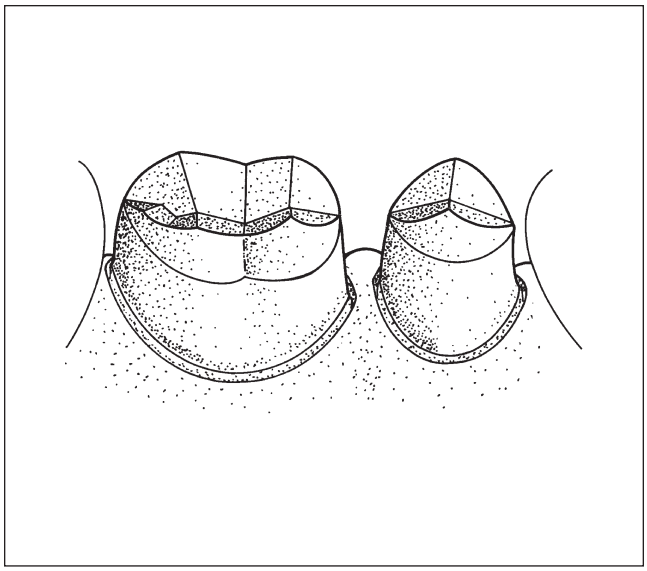


Fig 9-5 A full crown preparation is more retentive on a molar than on a premolar because the molar preparation has greater surface area.

Kent and associates[15] evaluated the degree of taper of 418 preparations, cut over 12 years by one operator. They found a mean of 15.8 degrees between mesial and distal walls and 13.4 degrees between facial and lingual walls for preparations in all areas of the mouth, with an overall mean of 14.3 degrees. The lowest combined taper (9.2 degrees) was seen on 145 anterior metal-ceramic crown preparations, while the greatest (22.2 degrees) was measured on 88 mandibular full crowns. Nordlander et al,[16] analyzing 208 preparations done by 10 dentists, reported a low of 17.3 degrees for premolars and a high of 27.3 degrees for molars, with an overall mean of 19.9 degrees.

Tooth preparation taper should be kept minimal because of its adverse effect on retention, but Mack[11] estimates that a minimum taper of 12 degrees is necessary just to ensure the absence of undercuts. The tendency to overtaper preparations is one that must be vigilantly guarded against in order to produce preparations with the least possible taper and the greatest possible retention. Consciously attempting to create a taper can easily result in an overtapered and nonretentive preparation. A taper or total convergence of 16 degrees has been proposed as being achievable clinically while still affording adequate retention.[17,18] This is probably an acceptable overall target. It can be as low as 10 degrees on preparations on anterior teeth and as high as 22 degrees on molars. Recommendations for degree of taper for specific teeth are given in Table 9-1.

Cement creates a weak bond, largely by mechanical interlocks, between the inner surface of the restoration and the axial wall of the preparation. Therefore, the greater the surface area of a preparation, the greater its retention.[5,19] Simply stated, preparations on large teeth are more retentive than preparations on small teeth (Fig 9-5). This is a factor that must be considered when a preparation is done on a small tooth, especially when it is an abutment for a fixed partial denture or a splint. Surface area can be increased somewhat by adding boxes and grooves. However, the benefits derived from such features may relate more to their limiting the freedom of movement than to the increase in surface area.

Freedom of displacement

Retention is improved by geometrically limiting the numbers of paths along which a restoration can be removed from the tooth preparation.[20] Maximum retention is achieved when there is only one path. A full veneer preparation with long, parallel axial walls and grooves would produce such retention (Fig 9-6a). On the opposite extreme, a short, overtapered preparation would be without retention because the restoration could be removed along an infinite number of paths (Fig 9-6b). The best preparation, then, is one that approaches the ideal and can be achieved within the limits of operator skill, accessibility, and laboratory technology.

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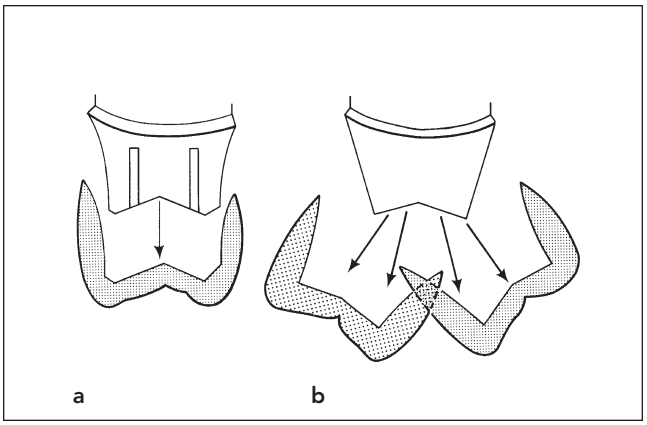


Fig 9-6 *(a)* By limiting the paths of withdrawal, retention is improved. *(b)* A preparation with unlimited freedom of displacement is much less retentive.

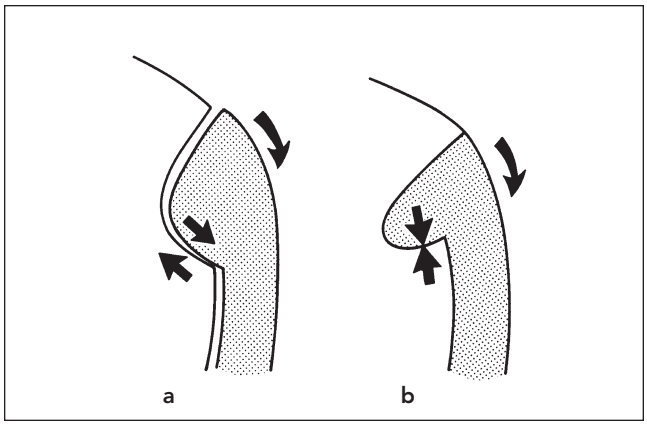


Fig 9-7 *(a)* The walls of a groove that meet the axial wall at an oblique angle do not provide the necessary resistance. *(b)* The walls of a groove must be perpendicular to rotating forces to resist displacement.

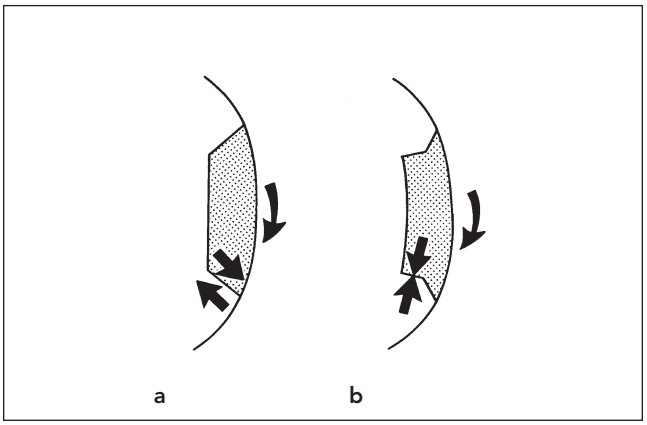


Fig 9-8 *(a)* The facial and lingual walls of a box will not resist rotational displacement if they form oblique angles with the pulpal wall. *(b)* They must meet the pulpal wall at angles near 90 degrees.

Limiting the freedom of displacement from torquing or twisting forces in a horizontal plane increases the resistance of a restoration. A groove whose walls meet the axial wall at an oblique angle does not provide the necessary resistance (Fig 9-7a). V-shaped grooves produce roughly one-half as much resistance to lingual displacement as do grooves with a definite lingual wall.[21] Forces that produce rotating movement in the restoration can produce shear and eventual slippage along the surfaces oblique to the direction of the force. There must be a definite wall perpendicular to the direction of the force to sufficiently limit the freedom of displacement and provide adequate resistance (Fig 9-7b).

A proximal box must be treated in a similar manner. If its facial and lingual walls form oblique angles with its pulpal wall, there will not be adequate resistance to rotating forces (Fig 9-8a). The facial and lingual walls must meet the pulpal wall at angles near 90 degrees so that these walls will be perpendicular to any forces that would tend to rotate the restoration (Fig 9-8b). A flare is then added to the box so that there can be an acute edge of gold at the cavosurface margin of the restoration.

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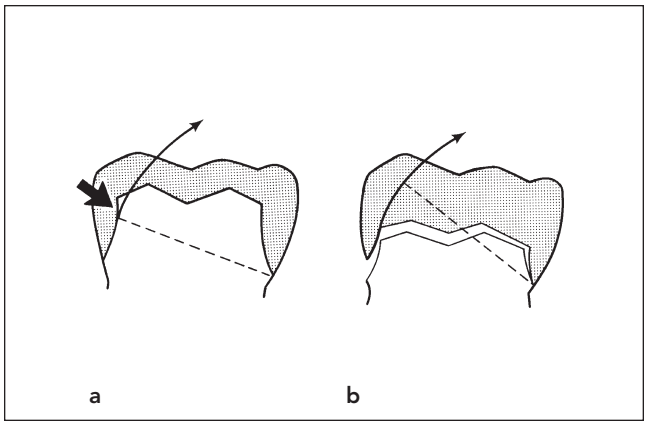


Fig 9-9 The preparation with longer walls *(a)* interferes with the tipping displacement of the restoration better than the short preparation *(b)*.

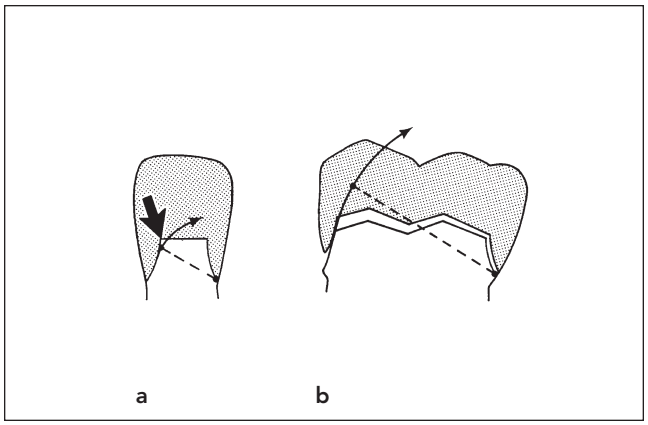


Fig 9-10 A preparation on a tooth with a smaller diameter *(a)* resists pivoting movements better than a preparation of equal length on a tooth of larger diameter *(b)*.

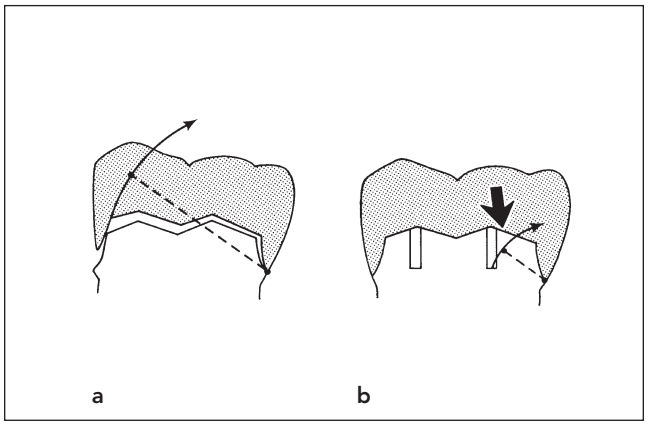


Fig 9-11 The resistance of a short preparation *(a)* can be improved by adding grooves *(b)*.

Length

Occlusogingival length is an important factor in both retention and resistance. Longer preparations will have more surface area and therefore will be more retentive. Because the axial wall occlusal to the finish line interferes with displacement, the length and inclination of that wall become factors in resistance to tipping forces.

For the restoration to succeed, the length must be great enough to interfere with the arc of the casting pivoting about a point on the margin on the opposite side of the restoration[22] (Fig 9-9a). The shorter wall does not afford this resistance (Fig 9-9b). The shorter the wall, the more important its inclination. The walls of shorter preparations should have as little taper as possible to increase the resistance. However, even this will not help if the walls are too short.

It may be possible to successfully restore a tooth with short walls if the tooth has a small diameter. The preparation on the smaller tooth will have a short rotational radius for the arc of displacement, and the incisal portion of the axial wall will resist displacement (Fig 9-10a). The longer rotational radius on the larger preparation allows for a more gradual arc of displacement, and the axial wall does not resist removal (Fig 9-10b). Parker et al[23] found that approximately 95% of anterior preparations analyzed had resistance form, while only 46% of those on molars did.

Resistance to displacement for a short-walled preparation on a large tooth can be improved by placing grooves in the axial walls. In effect, this reduces the rotational radius, and the portion of the walls of the grooves near the occlusal surface of the preparation will interfere with displacement (Fig 9-11).

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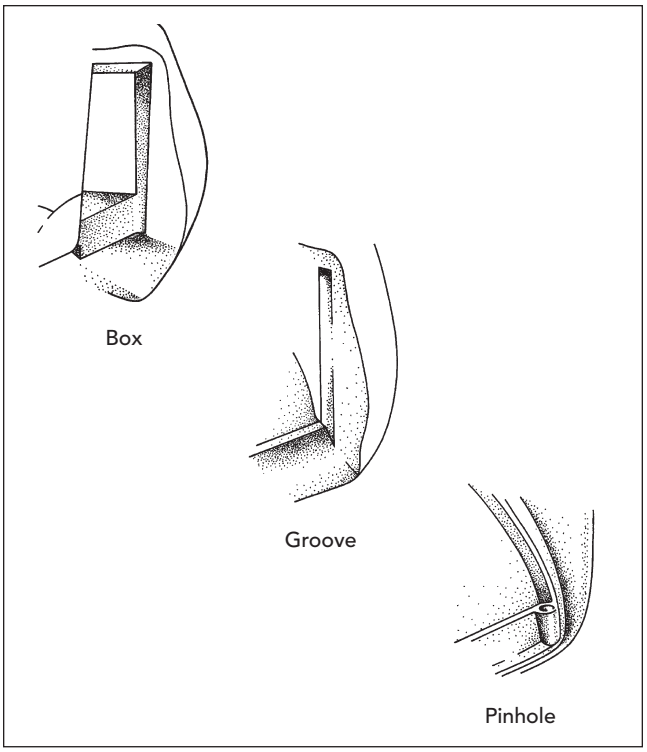


Fig 9-12 Internal preparation features such as the box, groove, and pinhole are frequently substituted for each other.

Substitution of internal features

The basic unit of retention for a cemented restoration is two opposing axial walls with a minimal taper. It may not always be possible to use opposing walls for retention: One may have been destroyed previously, or it may be desirable to leave a surface uncovered for a partial coverage restoration. It may also be that the walls are present but have a greater-than-desirable inclination. Generally, internal features such as the groove, the box form, and the pinhole are interchangeable and can be substituted for an axial wall or for each other (Fig 9-12). Substitution is important because conditions often preclude making an ideal preparation.

Kent et al[15] reported a marked difference between the degree of taper of full crown preparations (18.4 to 22.2 degrees) and that of boxes and grooves in the axial surfaces of those preparations (7.3 degrees). The taper of these internal features is nearly the same as the taper of the instruments used to cut them (4 to 6 degrees). Apparently the widely separated axial walls of the preparations are overinclined because of access, visibility, or both. In preparing an internal feature such as a groove or a box, however, the much shorter distance between the walls allows the dentist to prepare them more precisely. These features offer an excellent means of enhancing the overall retention and resistance of an otherwise overinclined axial wall. Woolsey and Matich[24] found that proximal grooves on short 15-degree dies provide complete resistance to faciolingual horizontal displacement.

Path of insertion

The *path of insertion* is an imaginary line along which the restoration will be placed onto or removed from the preparation. It is determined mentally by the dentist before the preparation is begun, and all features of the preparation are cut to coincide with that line. The path of insertion is not arbitrarily set at the completion of the preparation by adding a feature, such as grooves. It is of special importance when preparing teeth to be fixed partial denture abutments because the paths of all the abutment preparations must parallel each other.

The correct technique must be used to survey a preparation visually because this is the primary means of ensuring that the preparation is neither undercut nor overtapered. If the center of the occlusal surface of a preparation is viewed with one eye from a distance of approximately 30 cm (12 inches), it is possible to sight down the axial walls of a preparation with a minimum taper (Fig 9-13). However, it is also possible to sight down the axial walls of a preparation with a reverse (ie, undercut) taper of 8 degrees when both eyes are open (Fig 9-14). This occurs because of the distance between the eyes, which is responsible for binocular vision. Therefore, it is important that preparations be viewed with one eye closed.

For a preparation to be surveyed in the mouth, where direct vision is rarely possible, a mouth mirror is used (Fig 9-15). It is held at an angle approximately ½ inch above the preparation, and the image is viewed with one eye. If fixed partial denture abutment preparations are being evaluated for a common path of insertion, a firm finger rest is established, and the mirror is maneuvered until one preparation is centered. Then, pivoting on the finger rest, the mirror is moved, without changing its angulation, until it is centered over the second preparation.

The path of insertion must be considered in two dimensions: faciolingually and mesiodistally. The faciolingual orientation of the path can affect the esthetics of metal-ceramic or partial veneer crowns. For metal-ceramic crowns, the path is roughly parallel with the long axis of the teeth (Fig 9-16). A facially inclined path of insertion on a preparation for a metal-ceramic crown will leave the facio-occlusal angle too prominent, resulting in overcontouring of the restoration, opaque show-through, or both.

Leaning the path to the facial will force the overcutting of the mesiofacio-occlusal corner of a three-quarter crown preparation, leading to an unnecessary display of gold. For three-quarter crowns on anterior teeth, the path of insertion should parallel the incisal one-half to two-thirds of the labial surface (Fig 9-17). If it is inclined more facially, short grooves and an unnecessary display of gold will result.

The mesiodistal inclination of the path must parallel the contact areas of adjacent teeth. If the path is inclined mesially or distally, the restoration will be held up at the proximal contact areas (ie, locked out) (Fig 9-18). This is a particular problem when restoring a tilted tooth. In this situation, making the path of insertion parallel with the long axis of the tooth will cause the contacts of the adjacent teeth to encroach on the path of insertion.

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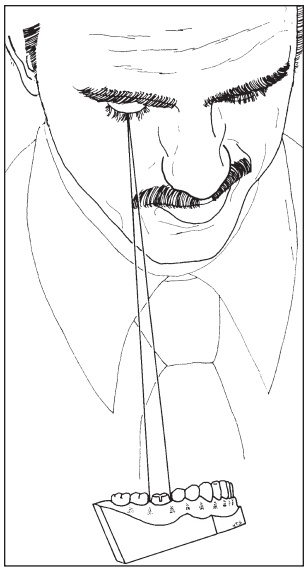


Fig 9-13 To examine a preparation for undercuts, one eye should be closed.

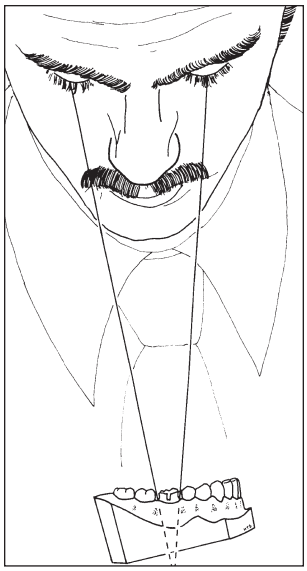


Fig 9-14 If both eyes are open when the preparation is viewed, undercuts may remain undetected.

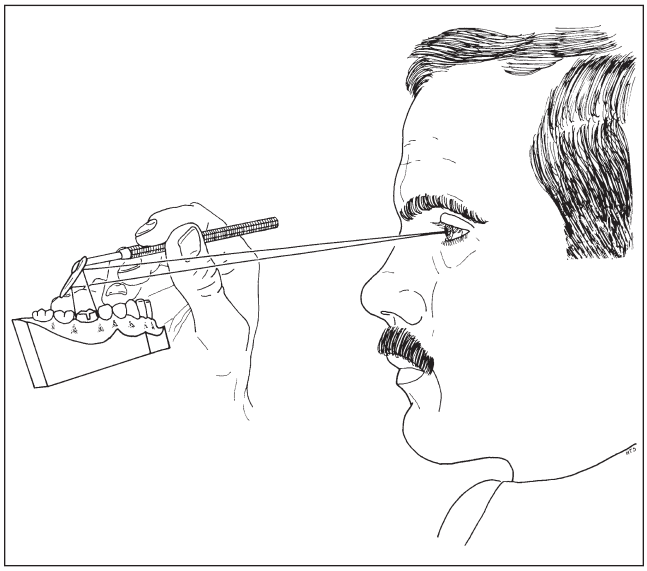


Fig 9-15 Preparations in the mouth are viewed through a mouth mirror using one eye.

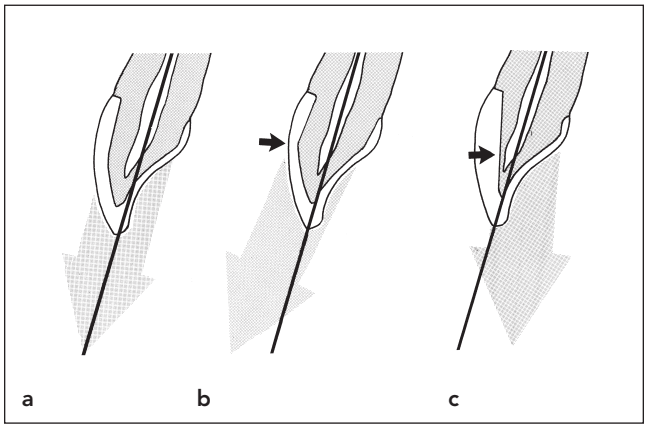


Fig 9-16 *(a)* The path of insertion of a preparation for a metal-ceramic crown should parallel the long axis of the tooth. *(b)* If the path is directed facially, the prominent facioincisal angle may create esthetic problems of overcontouring or opaque show-through. *(c)* However, if the path is directed lingually, the facial surface will intersect the lingual surface, creating a shorter preparation. It also may encroach on the pulp.

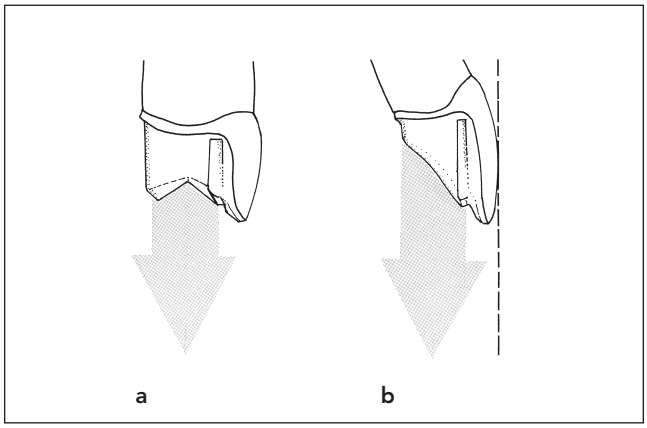


Fig 9-17 The path of insertion of a three-quarter crown on a posterior tooth parallels the long axis of the tooth *(a)*, whereas on an anterior tooth it parallels the incisal one-half to two-thirds of the labial surface *(b)*.

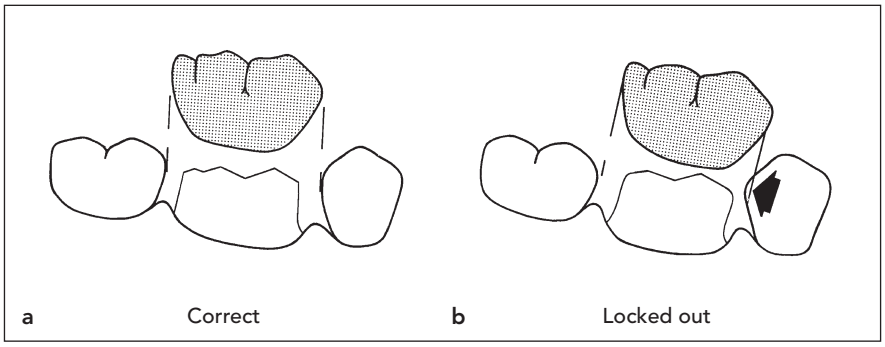


Fig 9-18 The path of insertion of a preparation must parallel the adjacent proximal contacts *(a)* or it will be prevented from seating *(b)*.

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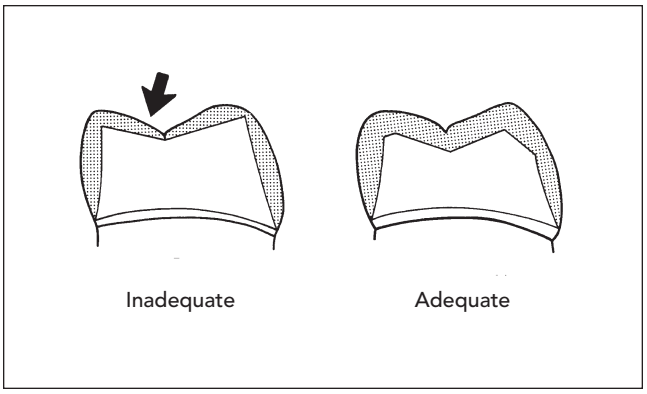


Fig 9-19 Inadequate occlusal reduction does not provide the needed space for a cast restoration of adequate thickness.

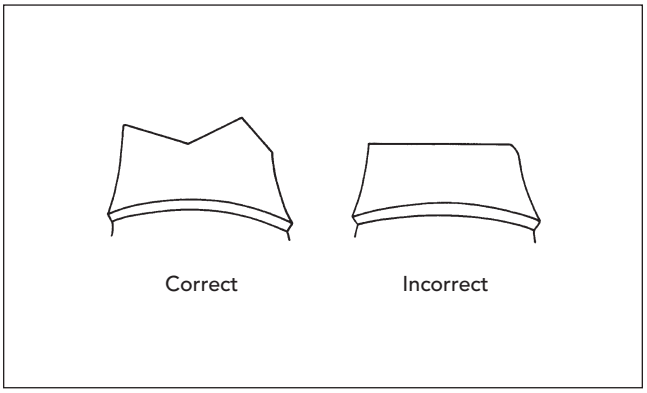


Fig 9-20 Occlusal reduction should reproduce basic inclined planes rather than being cut as one flat plane.

Structural Durability

A restoration must contain a bulk of material that is adequate to withstand the forces of occlusion. This bulk must be confined to the space created by the tooth preparation. Only in this way can the occlusion on the restoration be harmonious and the axial contours normal, preventing periodontal problems around the restoration.

Occlusal reduction

One of the most important features for providing adequate bulk of metal and strength to the restoration is occlusal clearance (Fig 9-19). For gold alloys, there should be 1.5 mm of clearance.

Metal-ceramic crowns will require 2.0 mm of clearance to receive ceramic coverage. There should be 2.0 mm of clearance on preparations for all-ceramic crowns. Malposed teeth may have occlusal surfaces that are not parallel with the occlusal table. Therefore, it may not be necessary to reduce the occlusal surface by 1.0 mm to achieve 1.0 mm of clearance.

The basic inclined plane pattern of the occlusal surface should be duplicated to produce adequate clearance without overshortening the preparation (Fig 9-20). A flat occlusal surface may overshorten a preparation that is already of minimal length to provide adequate retention. Inadequate clearance makes a restoration weaker. In addition, inadequate reduction under the anatomical grooves of the occlusal surface will not provide adequate space to allow good functional morphology. The restoration also will be much more easily perforated by finishing procedures or by wear in the mouth.

Functional cusp bevel

An integral part of the occlusal reduction is the functional cusp bevel (Fig 9-21). A wide bevel on the palatal inclines of the maxillary palatal cusps and the facial inclines of mandibular facial cusps provides space for an adequate bulk of metal in an area of heavy occlusal contact.

If a wide bevel is not placed on the functional cusp, several problems may occur. If the crown is waxed and cast to normal contour, the casting will be extremely thin in the area overlying the junction between the occlusal and axial reduction (Fig 9-22). To prevent a thin casting when there is no functional cusp bevel, an attempt may be made to wax the crown to optimal thickness in this area. An overcontoured restoration will result, and a deflective occlusal contact is likely to occur unless the opposing tooth is reduced (Fig 9-23).

If an attempt is made to obtain space for adequate bulk in a normally contoured casting without a bevel, the result will be an overcut axial surface (Fig 9-24). In addition to the unnecessary destruction of tooth structure, the severe inclination of the surface renders it useless for retention.

Axial reduction

Axial reduction also plays an important role in securing space for an adequate thickness of restorative material (Fig 9-25). If restorations are made with normal contours over preparations with inadequate axial reduction, they will have thin walls that will be subject to distortion. Laboratory technicians often attempt to compensate for this by overcontouring the axial surfaces. While this intended solution to the problem strengthens the restoration, it can have a disastrous effect on the periodontium.

There are other features that serve to provide space for metal that will improve the rigidity and durability of the restoration: the offset, the occlusal shoulder, the isthmus, the proximal groove, and the box (Fig 9-26). The isthmus connects the boxes, and the offset ties the grooves together to enhance the reinforcing "truss effect."[25]

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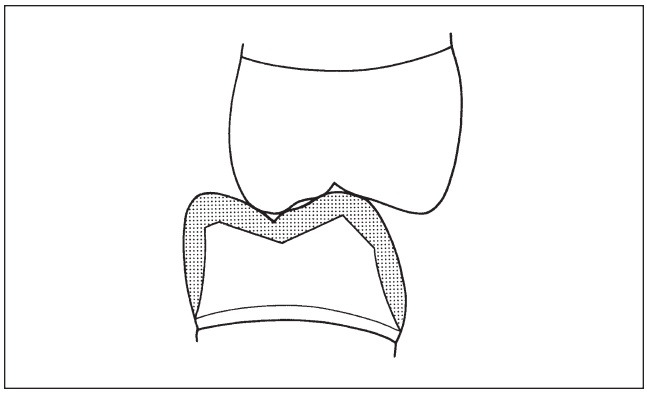


Fig 9-21 The functional cusp bevel is an integral part of occlusal reduction.

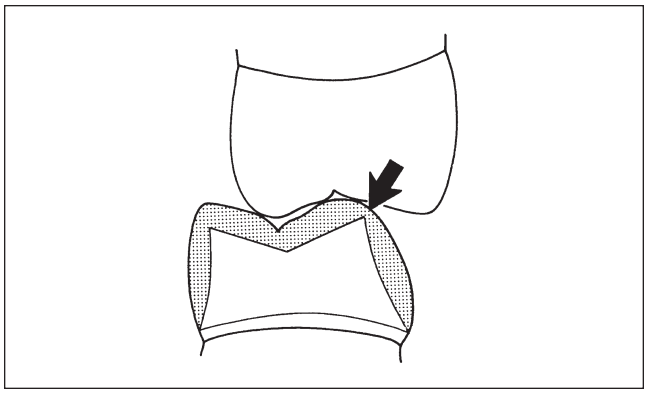


Fig 9-22 Lack of a functional cusp bevel can cause a thin area or perforation in the casting.

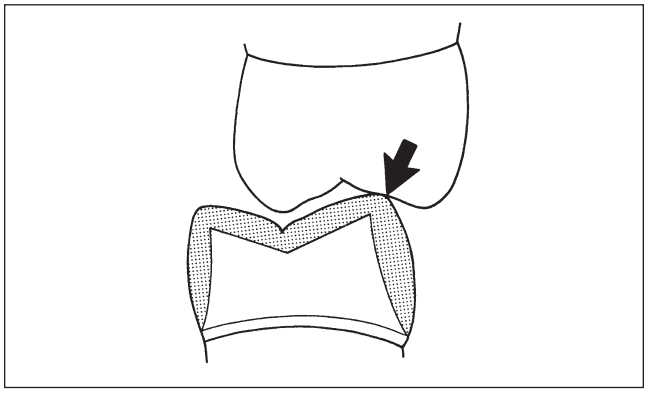


Fig 9-23 Lack of a functional cusp bevel may result in overcontouring and poor occlusion.

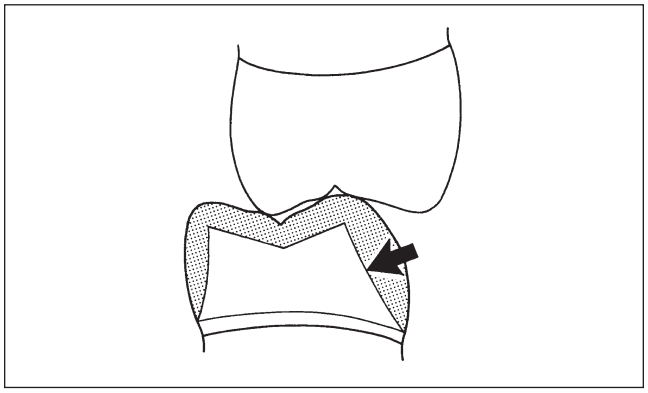


Fig 9-24 Overinclination of the facial surface will destroy excessive tooth structure and lessen retention.

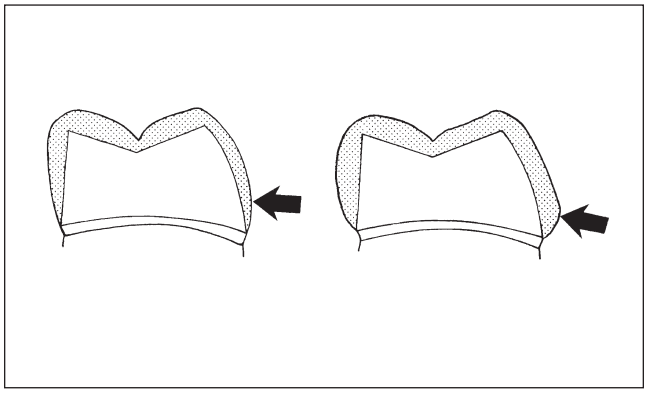


Fig 9-25 Inadequate axial reduction can cause thin walls and a weak restoration *(a)* or a bulbous, overcontoured restoration *(b)*.

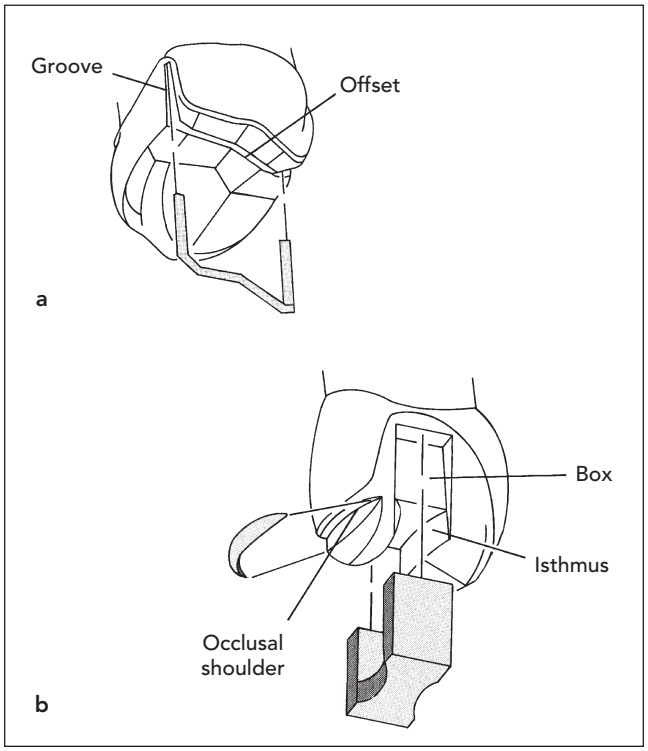


Fig 9-26 *(a)* The three-quarter crown is reinforced by the bulk of gold that fills the offset and grooves. *(b)* The occlusal shoulder strengthens the lingual margin, and the isthmus and boxes reinforce the main body of an MOD onlay.

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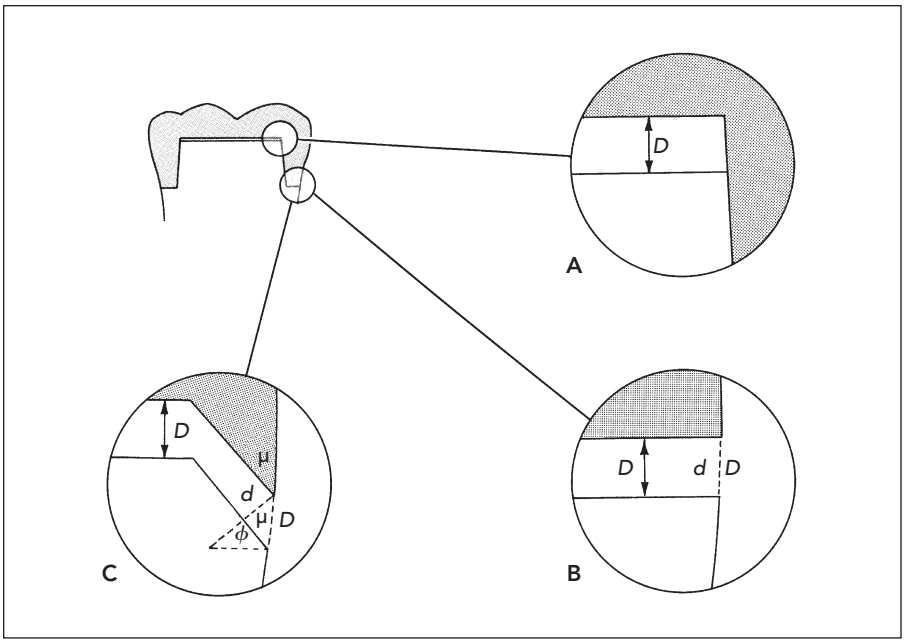


Fig 9-27 Any failure of the restoration to seat (*D* in inset A) is reflected as a marginal opening of the same dimension on a shoulder perpendicular to the path of insertion (inset B). As the angle of the margin (μ) approximates 0 degrees (inset C), the distance between the margin and the tooth (*d*) approaches 0, based on the assumption that the gap can be closed completely. (Modified from Rosner[26] with permission.)

Marginal Integrity

The restoration can survive in the biologic environment of the oral cavity only if the margins are closely adapted to the cavosurface finish line of the preparation. The configuration of the preparation finish line dictates the shape and bulk of restorative material in the margin of the restoration. It also can affect both marginal adaptation and the degree of seating of the restoration.

To bevel . . .

Cast metal restorations can be made to fit preparations with a high degree of precision, but even in well-fitting castings there is some discrepancy between the margin of the restoration and the preparation. Bevels have been advocated as a means of diminishing marginal discrepancy.[26] If the vertical discrepancy in fit is designated as *D*, the distance between the restoration and preparation (A in Fig 9-27) occurs unchanged between the margin and the finish line (B in Fig 9-27). However, the closest distance between the margin and the surface of the preparation is a line, *d*, that is perpendicular to the surface of the tooth (C in Fig 9-27). It can be stated as a function of *D* and the sine of angle μ or the cosine of angle ϕ:

$$d = D \sin \mu$$
or
$$d = D \cos \phi$$

Table 9-2 Trigonometric functions of angles 0 to 90 degrees

Angle (degrees)	Sine	Cosine
0	0	1.000
15	0.259	0.966
30	0.500	0.866
45	0.707	0.707
60	0.866	0.500
75	0.966	0.259
90	1.000	0

As angle μ becomes smaller (more acute), the sine of μ becomes smaller (Table 9-2); as angle ϕ becomes larger (more obtuse), the cosine of ϕ becomes smaller. By either computation, *d* diminishes by the same amount. The more acute the angle of the margin (μ) or the more obtuse the angle of the finish line (ϕ), the shorter the distance between the restoration margin and the tooth. This argument is based on the premise that the distance between the margin and tooth structure is infinitely closeable, and as long as there is no cement between the restoration and the preparation, that is true.

. . . Or not to bevel

However, as shown so convincingly by Ostlund,[27] the presence of cement changes the scenario completely. The film thickness of the cement will prevent the complete seating

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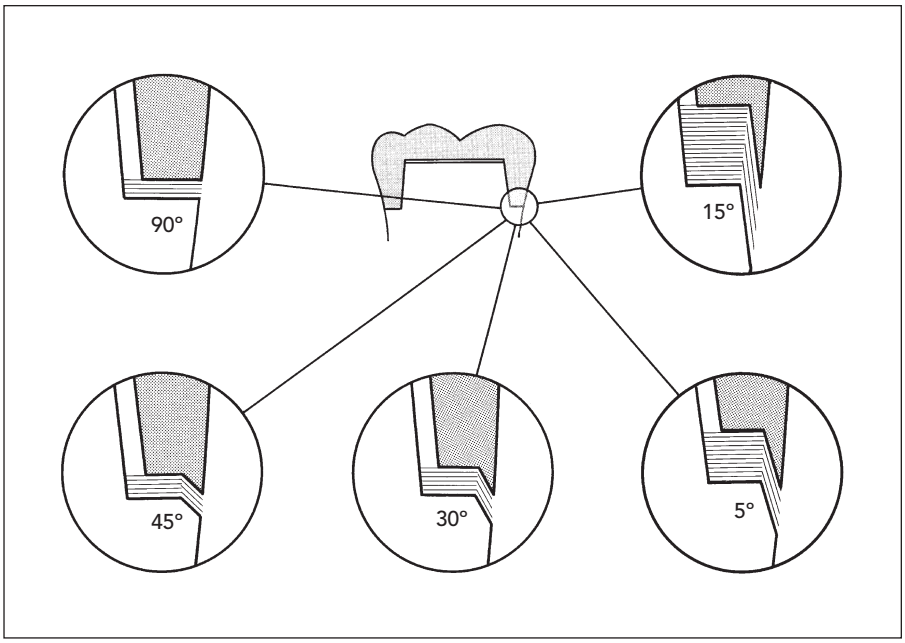


Fig 9-28 The cement film thickness prevents complete closure of the marginal gap. If a bevel of 45 degrees is added to a shoulder, the crown will be prevented from seating by a factor of 1.4. However, as the margin is decreased to an angle of 30 degrees, the crown is displaced twice as much as it would be with a shoulder. Margins of 15 and 5 degrees would prevent seating by factors of 3.9 and 11.5, respectively. If the marginal gap for the shoulder were 25 μm, the American Dental Association specification for cement film thickness, the addition of a 5-degree bevel could keep the casting from seating by nearly 0.3 mm. All spaces in the insets are drawn to scale.

of a casting with bevels that are nearly parallel with the path of insertion of the restoration, just as Jorgensen,[4] Kaufman et al,[5] and Eames and associates[14] found that crowns did not seat completely on dies with minimal taper.

The film thickness of the cement imposes a limit on the reduction of the perpendicular distance from the margin to the tooth, *d*. This distance therefore becomes a constant, and the previous equation is solved for *D* instead of *d*:

$$D = d/\sin \mu$$
or
$$D = d/\cos \phi$$

As the angle of the margin bevel becomes more acute, its sine becomes smaller, and as the angle of the finish line becomes more obtuse, its cosine becomes smaller, and D becomes larger. The more nearly the bevel parallels the path of insertion, the greater the distance by which the restoration fails to seat (Fig 9-28).

McLean and Wilson[28] have disputed the use of bevels for metal-ceramic crowns because the bevel margin must be 10 to 20 degrees to noticeably improve adaptation. The finish line must also be placed too far subgingivally to hide the resultant metal collar. Pascoe[29] demonstrated that slightly oversized castings with shoulders exhibit the least marginal discrepancy. Gavelis et al[30] found a better marginal seal with acute-edged margins, but they found that shoulders permitted the most complete seating of a crown. Panno and associates[31] reported no better adaptability of crowns with highly acute 80-degree bevels than those with less acute 45-degree bevels.

Empirical clinical results dictate that the acute angle margin should continue to be used on metal restorations but that the angle should be in the 30- to 45-degree range. The tapered edge in a wax pattern margin produced by a bevel is more readily adapted to a die than is a butt joint, and a gold margin can be burnished to slightly improve its adaptation after casting.

Finish line configurations

Wide, shallow bevels that are nearly parallel with the outer surface of the tooth should be avoided. They are likely to lead to overcontouring. Even if the axial surfaces of the overlying crown are not overcontoured, the resultant thin, unsupported wax at the margin potentially will break or distort when the wax pattern is withdrawn from the die and invested. The optimum margin for a gold alloy casting is an acute edge with a nearby bulk of metal.

The preferred gingival finish line for veneer metal restorations is the chamfer (Fig 9-29). This finish line has been shown experimentally to exhibit the least stress, so the cement underlying it will have less likelihood of failure.[32,33] It can be cut with the tip of a round-end diamond while the axial reduction is being done with the side of that instrument. However, a torpedo diamond is less likely to produce a butt joint. The margin of the cast restoration that fits against it combines an acute edge with a nearby bulk of metal.

A deep chamfer (also known as a *heavy chamfer*) is used to provide a 90-degree cavosurface angle with a large-radius rounded internal angle (Fig 9-30). The radius of curvature

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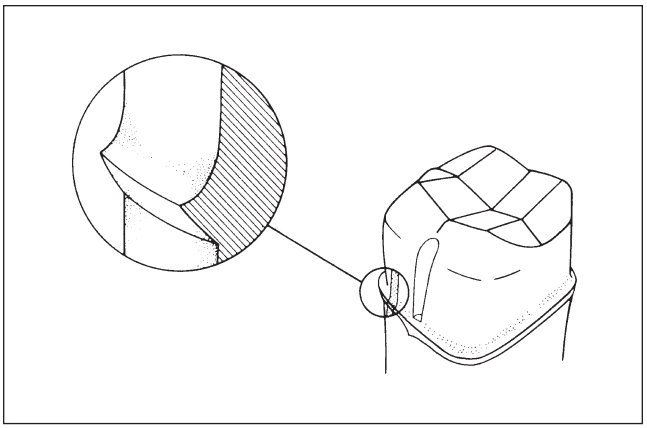


Fig 9-29 Chamfer finish line demonstrated on a full veneer crown preparation.

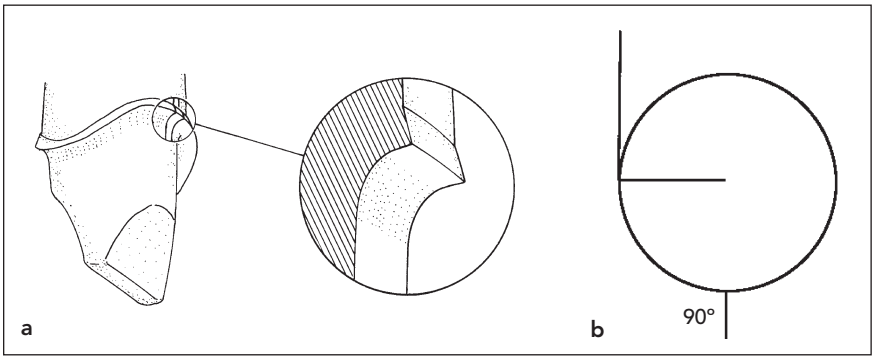


Fig 9-30 *(a)* Deep chamfer on a preparation for an all-ceramic crown. *(b)* Ninety-degree finish line formation.

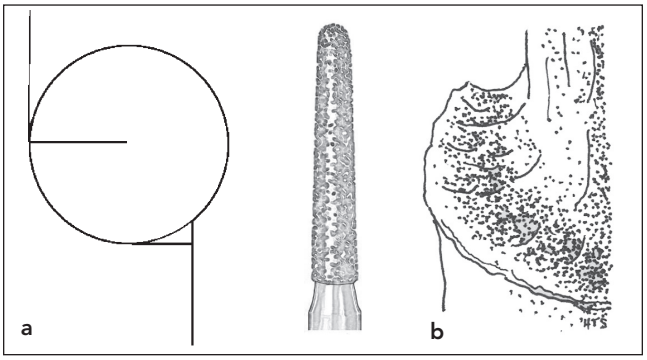


Fig 9-31 *(a and b)* Lip formation.

equals the depth of axial reduction. It is created with a round-end tapered diamond, which, in the hands of an unskilled operator, can create an undesirable fragile lip of enamel at the cavosurface (Fig 9-31). This friable, unsupported enamel is very easily fractured during or after cementation of the restoration. The deep chamfer provides better support for a ceramic crown than does a conventional chamfer, but it is not as good as a shoulder. A bevel can be added to the deep chamfer for use with a metal restoration.

The classic shoulder was long the finish line of choice for the all-ceramic crown (Fig 9-32). The wide ledge provides resistance to occlusal forces and minimizes stresses that might lead to fracture of the porcelain. It produces the space for healthy restoration contours and maximum esthetics. However, it does require the destruction of more tooth structure than any other finish line. The sharp, 90-degree internal line angle associated with the classic variety of this finish line concentrates stress in the tooth and is conducive to coronal fracture. The shoulder generally is not used as a finish line for cast metal restorations.

The radial shoulder is a modified form of shoulder finish line (Fig 9-33a). The initial instrumentation of the ledge is accomplished with the coarse, flat-end tapered diamond. A small-radius rounded internal angle is instrumented with the

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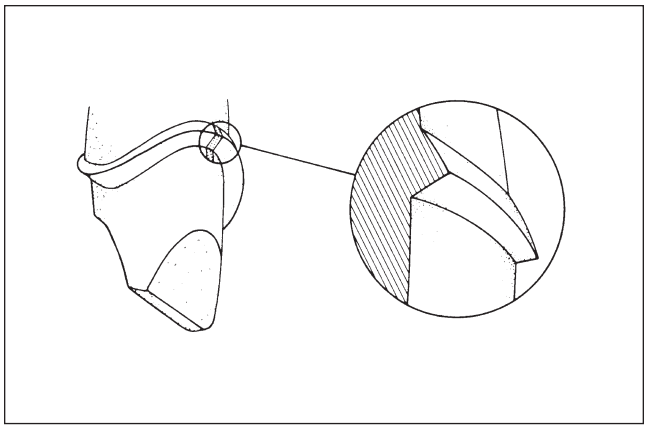


Fig 9-32 Classic shoulder on a preparation for an all-ceramic crown (a traditional porcelain jacket crown).

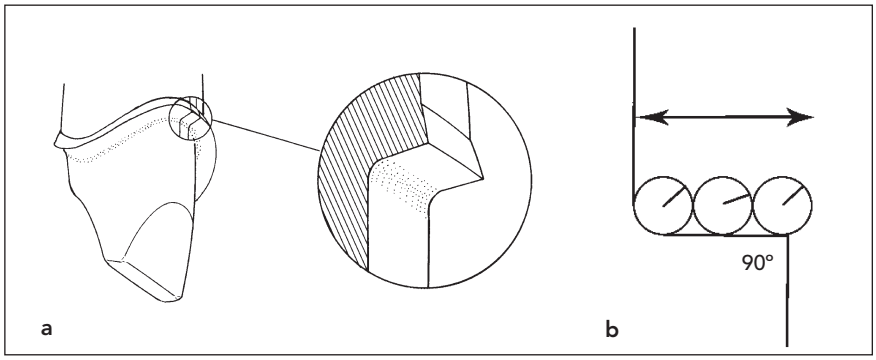


Fig 9-33 *(a)* A radial shoulder on an all-ceramic crown preparation combines maximum support of the ceramic with a stress-reducing, rounded gingivoaxial angle. *(b)* Minimal radius of curvature reduces stress.

Table 9-3 Finite element analysis of preparation finish lines*

Finish line	Stress (psi)	Reduction (%)	Change (%)
Classic shoulder	2,885	NA	NA
Radial shoulder	2,101	27	27
Deep chamfer	1,897	34	7

NA, not applicable.
*Based on data from Mullasseril.[34]

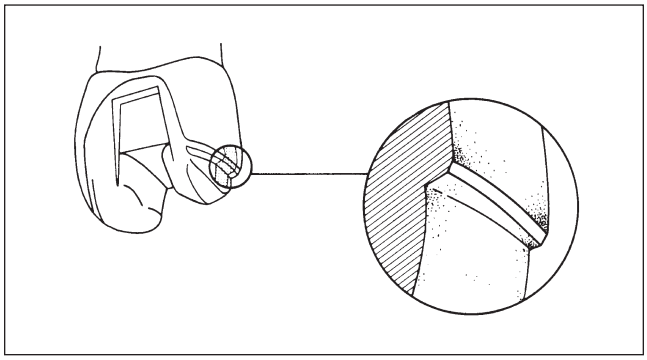


Fig 9-34 Shoulder with a bevel on the occlusal shoulder of an MOD onlay.

fine, flat-end tapered diamond, and finishing is completed with a specially modified binangle chisel. The cavosurface angle is 90 degrees, and shoulder width is only slightly lessened by the rounded internal angle. The radius of curvature equals one-fourth to one-fifth the depth of the axial reduction (Fig 9-33b). Stress concentration is less in the tooth structure than with a classic shoulder, and support for ceramic restoration walls is good. The destruction of tooth structure required for this configuration is not significantly less than that required for a classic shoulder, however. While a deep chamfer will produce slightly less stress than a radial shoulder, it only amounts to a 7% decrease[34] (Table 9-3).

The shoulder with a bevel is used as a finish line in a variety of situations (Fig 9-34). It is utilized as the gingival finish line on the proximal box of inlays and onlays and for the occlusal shoulder of onlays and mandibular three-quarter crowns. This design can also be used for the facial finish line of metal-ceramic restorations where gingival esthetics is not critical. It can be used in those situations where a shoulder is already present either because of destruction by caries or the presence of previous restorations. It is also a good finish line for preparations with extremely short walls because it facilitates axial walls that are nearly parallel.[35]

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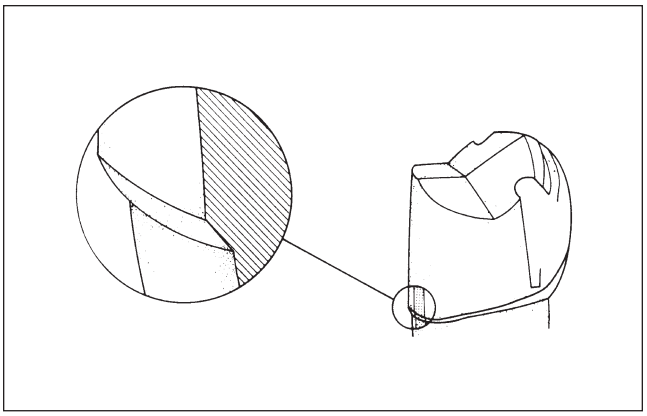


Fig 9-35 Knife edge on the lingual of a mandibular three-quarter crown.

Table 9-4 Advantages and disadvantages of finish lines

Finish line	Advantages	Disadvantages
Chamfer	Minimal tooth destruction Minimal stress	Reduces crown strength (ceramic) Poor esthetics (ceramic)
Deep chamfer	Moderate tooth destruction Minimal stress in tooth	Reduces crown strength Potential lip formation
Classic shoulder	Maximal esthetics Maximal crown strength Prevents overcontouring	Maximal destruction Maximal tooth stress
Radial shoulder	Maximal esthetics Excellent crown strength Less stress than classic shoulder	Destructive of tooth More stress than chamfer
Radial shoulder with a bevel	Excellent crown strength Less stress than classic shoulder	Destructive of tooth More stress than chamfer Poor esthetics (necessitates metal collar)
Knife edge	Minimal destruction	Overcontouring (ceramic) Poor esthetics Weaker crown margin

By adding a bevel to an existing shoulder, it is possible to create an acute edge of metal at the margin. The shoulder with a bevel should not be used routinely for full veneer restorations because the axial reduction required to obtain it is unnecessarily destructive of tooth structure. Some variation of a shoulder, with or without a bevel, may afford some resistance against distortion during porcelain firing.[36]

The ultimate in finish lines that permit an acute margin of metal is the *knife edge* (Fig 9-35). Unfortunately, its use can create problems. Unless it is carefully cut, the axial reduction may fade out instead of terminating in a definite finish line. The thin margin of the restoration that fits this finish line may be difficult to accurately wax and cast. It is also more susceptible to distortion in the mouth when the casting is subjected to occlusal forces.

The use of the knife edge can result in overcontoured restorations when an attempt is made to obtain adequate bulk by adding to the external axial contours of the restoration. In spite of its drawbacks, it is sometimes necessary to use the knife edge. It may have to be used on the lingual surface of mandibular posterior teeth, on teeth with very convex axial surfaces, and on the surface toward which a tooth may have tilted. The advantages and disadvantages of these finish line choices are summarized in Table 9-4.

The finish line used for the facio-occlusal margin of maxillary partial coverage and MOD onlay restorations is worthy of attention. It, too, must meet the requirement of providing an acute edge with a nearby bulk of metal. The enamel must also be protected by a finishing bevel that will leave the tooth structure at the cavosurface angle with sufficient

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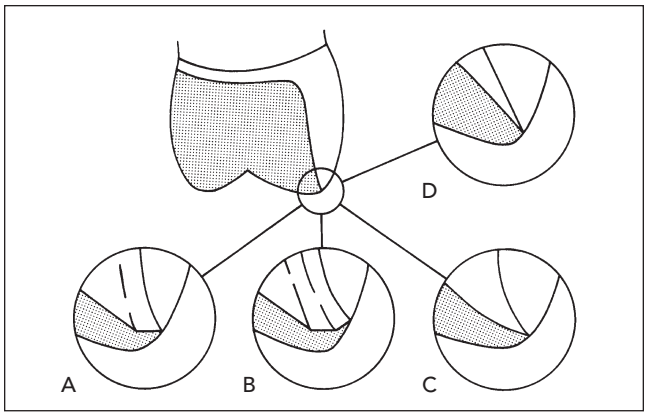


Fig 9-36 Facio-occlusal finish lines on a maxillary three-quarter crown. A flat bevel (A), a contrabevel (B), and, when the cusp is bulky, a knife edge (C), are acceptable finish lines. The knife edge is not an acceptable finish line on small, sharp cusps (D). (Adapted from Ingraham et al[37] and Richter and Ueno.[38])

bulk to resist fracture and chipping.[37] The most commonly used form is a narrow (0.3- to 0.5-mm) finishing bevel perpendicular to the path of insertion of the restoration (A in Fig 9-36). A contrabevel may also be used where function is heavy and esthetic requirements are minimal (B in Fig 9-36). There are a few situations in which no bevel is required (C in Fig 9-36), but this can only be accomplished on a cusp that is bulky enough to allow the acute edge of metal and still finish the enamel at the cavosurface angle. A bevel is mandatory if its elimination will create an unsupported edge of enamel (D in Fig 9-36).

Preservation of the Periodontium

The placement of finish lines has a direct bearing on the ease of fabricating a restoration and on the ultimate success of the restoration. The best results can be expected from margins that are as smooth as possible and are fully exposed to cleansing.[39] Whenever possible, the finish line should be placed in an area where the margins of the restoration can be finished by the dentist and kept clean by the patient. In addition, finish lines must be placed so that they can be duplicated by the impression, without tearing or deforming the impression when it is removed past them.

Finish lines should be placed in enamel when it is possible to do so. In the past, the traditional concept was to place margins as far subgingivally as possible, based on the mistaken concept that the subgingival sulcus is caries free.[40] The practice of routinely placing margins subgingivally is no longer acceptable. Subgingival restorations have been described as a major etiologic factor in periodontitis.[41–48] The deeper the restoration margin resides in the gingival sulcus, the greater the inflammatory response.[49–52]

Although Richter and Ueno[38] reported no difference between subgingival and supragingival margins in a 3-year clinical study, they recommended that placement be supragingival whenever possible. Eissmann et al[39] made a similar recommendation. Koth[53] also failed to find a link between margin location and gingival health in a selected patient population on a strict hygiene regimen.

These studies do not refute the evidence that subgingival margins are likely to cause gingival inflammation. They merely demonstrate that margin location is not as crucial when placed by a highly skilled dentist in the mouth of a motivated, cooperative patient. Ego may tempt a dentist to believe that he or she has the skills to achieve well-fitting subgingival margins on a crown. However, subgingival margins can be very difficult to evaluate.

Christensen[54] demonstrated that experienced restorative dentists can miss marginal defects as great as 120 µm when the margins are subgingival. In a radiographic study, Bjorn et al[55] found that more than half of the proximal margins of gold crowns had defects greater than 0.2 mm and more than 40% of the proximal margins of ceramic crowns had defects that exceeded 0.3 mm.

Nonetheless, there will be many situations in which subgingival margins are unavoidable. Because preparation length is such an important factor in resistance and retention, preparations are frequently extended subgingivally to increase retention.[47,56–59] The placement of finish lines can also be altered from ideal locations by caries,[47,56–59] the extensions of previous restorations,[47,56–58] trauma,[47,58] or esthetics.[47,56–59]

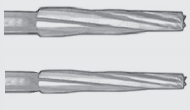
Caution should be exercised if conditions require that the finish line be placed any closer to the alveolar crest than 2.0 mm, which is the combined dimension of the epithelial and connective tissue attachments.[60] Placement of a restoration margin in this area probably will result in gingival inflammation, loss of alveolar crest height, and formation of a periodontal pocket.[61] Crown lengthening may be performed to surgically move the alveolar crest 3.0 mm apical to the location of the proposed finish line to guarantee the biologic width and prevent periodontal pathology. This will allow space for the connective and epithelial attachments and a healthy gingival sulcus. If the deep finish line is located interproximally and will require extensive removal of bone between the tooth being restored and the adjacent tooth, it may be better to extract the tooth in question rather than periodontally compromise its healthy neighbor.

Instrumentation

The preparation of teeth to receive cast metal or ceramic restorations does not require an extensive armamentarium (Table 9-5).

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Table 9-5 Rotary instruments for tooth preparations

Shape	ISO no.	Silhouette	Use
Round-end tapered diamond	6856-016		Depth-orientation grooves Occlusal reduction Functional cusp bevel
Round-end tapered finishing diamond	8856-016		Occlusal finishing Functional cusp bevel finishing
Flat-end tapered diamond	6847-016		Axial reduction (MCR, all-ceramic crown) Shoulder (MCR, all-ceramic crown)
Flat-end tapered finishing diamond	8847KR-016		Axial wall finishing (MCR, all-ceramic crown) Radial shoulder (MCR, all-ceramic crown)
Tapered torpedo diamond	6877K-014		Axial reduction Chamfer finish line
Tapered torpedo finishing diamond	8877K-014		Axial wall finishing Chamfer finishing
Short needle diamond	852-012		Initial proximal reduction (posterior teeth)
Long needle diamond	850-012		Initial proximal reduction (anterior teeth)
Small football-shaped diamond	6379-021		Lingual reduction (anterior teeth)
Small football-shaped finishing diamond	8379-021		Lingual reduction finishing (anterior teeth)
Flame diamond	862-010		Proximal flare Gingival bevel
Flame finishing diamond	8862-010		Flare and bevel finishing
Tapered fissure bur	171L-012		Seating groove Proximal groove (posterior teeth) Offset Occlusal shoulder Isthmus Proximal box Smoothing and finishing Occlusal and incisal bevels
Tapered groove alignment fissure burs	170L-010 169L-009		Initial groove alignment Angles of proximal boxes Smoothing and finishing Occlusal and incisal bevels

MCR, metal-ceramic restoration.

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The excavation of caries should be accomplished with sharp spoon excavators and round burs (no. 4 or no. 6) mounted in a contra-angle handpiece. Hand chisels may be used to accentuate the facial and lingual walls of proximal boxes. All other procedures usually are accomplished with a high-speed air turbine handpiece.

Small diamond points, used with an air-water spray in a high-speed handpiece, will remove precisely controlled amounts of tooth structure. The surface that remains can be easily smoothed. There is no indication for the use of large diamond cutting disks in low-speed contra-angle or straight handpieces. They frequently overextend preparations, and their potential for injury to the patient is great.

It is important that the cavosurface finish line be smooth and continuous to facilitate the fabrication of restorations with well-adapted margins. Gross reduction is most efficiently accomplished with coarse diamonds. However, they leave irregular cavosurface finish lines[62,63] and should not be used to obtain a smooth finish line. Using fine diamonds of the same size and shape, it is possible to maintain the finish line configuration developed by the coarse diamond instrument. Acceptable finish lines on vertical flares can be obtained through the use of abrasive paper disks,[64] but they should be used with a rubber dam to protect soft tissue.

Nondentate tapered burs (169L, 170L, and 171L) are used for grooves, boxes, isthmuses, and offsets where they are needed. They are also used for smoothing any surface that will not terminate in a curved finish line, which they would nick, and for creating occlusal and incisal bevels. Cross-cut or dentate burs are employed for removal of old restorations, but the horizontal ridges they leave on tooth structure make them unacceptable for planing tooth surfaces.

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